

1 Few Essential Rules

- The assignments should be done in a group of two students.
- At least one of the team members must submit the assignment.
- All submissions will be checked for plagiarism.
- Discussions are allowed and encouraged.
- While working in groups, try using a version control system like GitHub.

2 Assignment Statement

- Go to the website NYU Depth V2.
- Download Labeled dataset (~ 2.8 GB).
- Open the dataset using MATLAB (the file format is in .mat !!).
- There are two datasets in matrix format: images ($H \times W \times 3 \times N$) and depths ($H \times W \times N$). Save them in an external file and read them in your Python code.
- Take any one image and its corresponding depthmap (Z_d).
- These depths are projected to the RGB image plane associated with the RGB camera.

- Use the pinhole camera parameters model to estimate the X_{rgb} and Y_{rgb} .

$$\left[X_{rgb} = \frac{(x_{rgb} - c_{x_{rgb}})Z_d}{f_{x_{rgb}}}, Y_{rgb} = \frac{(y_{rgb} - c_{y_{rgb}})Z_d}{f_{y_{rgb}}} \right] \text{ where}$$
$$f_{x_{rgb}} = 5.1885790117450188e+02, f_{y_{rgb}} = 5.1946961112127485e+02, c_{x_{rgb}} = 3.2558244941119034e+02, c_{y_{rgb}} = 2.5373616633400465e+02;$$

- Now, you have a one-to-one correspondence with the pixel values of the RGB image with the real-world coordinate points. Pick up $n \geq 6$ number of such correspondence points.
- Finally, implement the linear camera calibration approach from section 1.3.1 of Ponce and Forsyth (explained in the class also) to find out the whole perspective projective matrix $\mathcal{M}$.

3 Submission Details

Each submission must include a report containing the details of all the team members, the working principle of your codebase, and important tables of the output results. Additionally, it is recommended that you briefly explain your code and implementation details in the report.